MOSCOW EXCHANGE STOCKS POPULARITY TRACKER

by

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STARTUP AS A THESIS

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ABSTRACT

This thesis project endeavours to establish a profitable small IT-business in the InvestTech (Investment Technologies) domain booming in Russia nowadays. The product is a Moscow Exchange Stocks Popularity Tracker tailored for retail investors. The methodology employed incorporates principles from Lean Startup, Agile, emphasising efficiency, fast testing of hypotheses and resource optimization. The result of the thesis shows that minimal viable product (MVP) and first sales can be made hardly without any investments.

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1. INTRODUCTION

This thesis chronicles the journey of the Russian Stock Popularity Tracker, from its inception as a mere concept to its realization as a startup. Through the lens of the "startup as a thesis" format, we go into the process of building and validating this venture, exploring key methodologies, market research endeavors, strategies for achieving product-market fit, implementation efforts, and in-depth analysis and discussions.

Chapter 2 lays the foundation by examining the design and methodology employed in the development of the Russian Stock Popularity Tracker. From bootstrapping and the utilization of state grants to the application of Lean Startup, Agile methodologies, and the HADI Cycles framework, this chapter elucidates the approach taken in shaping the startup's trajectory.

In Chapter 3, we embark on a journey into market research, delving into the nuances of the industry, analyzing market size, and scrutinizing the competitive landscape to glean insights that inform the startup's strategy and direction.

Chapter 4 dives deep into the pursuit of achieving product-market fit, elucidating the methodologies and frameworks utilized to understand customer needs, identify solutions, and map out the customer journey to optimize engagement and conversion.

The subsequent chapters, Chapter 5 and Chapter 6, navigate the implementation phase and go into an analysis and discussion of the startup's results, respectively. From the development of a minimal viable product (MVP), tracking key metrics, analyzing unit economics, and exploring scalability considerations to profit & loss projections and captable, these chapters provide a comprehensive overview of the startup's journey.

Finally, Chapter 7 encapsulates the conclusions drawn from the research and offers reflections on the lessons learned, and the future prospects of the Russian Stock Popularity Tracker. Through this exploration, this thesis aims to shed light on the process of building and validating a startup, offering insights, lessons, and inspiration for aspiring entrepreneurs embarking on their own entrepreneurial endeavors.

2. DESIGN AND METHODOLOGY

Probability of launching a successful startup and transforming it into a sustainable business is extremely low. Statistically speaking, launching a startup is a bad idea. The goal I follow while writing this chapter is to describe practices we follow to increase chances of success.

First, let’s look at top 2 reasons why startups fail according to many sources:

1. **Insufficient funding**: Cash flow problems are a leading cause of startup failure. Whether due to underestimating costs, poor financial management, or an inability to secure additional funding, running out of money can quickly derail a startup's operations.
2. **Lack of product market fit**: Many startups fail because they create a product or service that doesn't solve a significant problem or meet a real need in the market. Without demand, even the most innovative ideas can struggle to gain traction.

In this chapter I will describe methods we choose in order to mitigate both of these major risks.

2.1. BOOTSTRAPPING AND STATE GRANTS

In order to mitigate the risk of failure because of insufficient funding, the chosen approach is **bootstrapping** in the team of two:

- Founder – the specialist with experience in product marketing and product management
- Partner (the full stack developer).

This team stack is enough to launch the first version of a product and attract first sales and then prove unit economics of the startup. At this moment there will be 3 options:

- To attract funding on beneficial terms and to enlarge product team;
- To pivot because the product is proved to be a failure;
- To keep the product as a pet project and continue to bootstrap.

In order to cover additional costs, state funding by fasie.ru is attracted. Such additional costs can be: server, first user acquisition campaigns, design etc. The grant is 1 million roubles.
State funding has its pros and cons. Pro is that you have a high probability to attract funding and have no risk of losing control of your company. Con is that the process is quite long, bureaucratic and gives you significant risks in case of pivots. The reason for this risk is that the government gives you money for a specific idea, and if you pivot there is no guarantee that they will not ask for a refund.
We chose to go for a state grant because we were introduced to this mechanism by the Innopolis University's Startup Studio and were led through all the steps.

2.2. LEAN STARTUP

Bootstrapping is indeed a way to mitigate the risk of insufficient funding, as it drastically reduces costs involved in the process. However, the lifespan of a team, which adopts bootstrapping, is still limited. It means that we need to speed up the process of product development as much as possible. In order to do that the **lean startup methodology** was chosen.

The Lean Startup methodology is an approach to developing businesses and products, which was popularized by entrepreneur Eric Ries in his book "The

Lean Startup." It emphasizes the importance of rapidly iterating on ideas, gathering customer feedback, and making data-driven decisions to build successful and sustainable businesses.

Three pillars of lean startup are following:

- Business Model Canvas or Value Proposition Canvas (Learn)
- Minimum Viable Product (Build)
- Customer development (Measure)

Overall, the Lean Startup methodology is designed to help startups build products that customers actually want, while minimizing wasted time, resources, and effort through rapid experimentation and continuous learning. It's about embracing uncertainty, taking calculated risks, and focusing on delivering value to customers.

2.3. AGILE

Another methodology we adopted while developing this product is Agile. While Lean Startup is an approach of doing early-stage IT-business, agile is a methodology of software development, which is only a part of doing IT-business.

The reason we adopted agile is that we need to communicate clearly between partners and freelance specialists on how we build our product and rapidly adapt to gathered information during the customer development research, which we use as a part of the lean startup methodology. The agile manifesto with it’s 4 principles allow to shape work structure before you start:

- Individuals and interactions over processes and tools

- Working software over comprehensive documentation
- Customer collaboration over contract negotiation
- Responding to change over following a plan

2.3. HADI CYCLES

Hadi Cycles, also known as HADI (Hypothesis, Action, Data, Insights) framework, is a methodology used in startup building to guide iterative experimentation and decision-making. It provides a structured approach for startups to validate hypotheses, take action based on insights gained from data, and continuously improve their strategies and products. Here's how Hadi Cycles are typically used in startup building:

1. **Hypothesis:** Startups begin by formulating hypotheses about various aspects of their business, such as customer needs, market demand, product features, or marketing strategies. These hypotheses are educated guesses based on assumptions and prior knowledge, and they serve as the foundation for experimentation.
2. **Action:** Once hypotheses are formulated, startups take action to test them. This may involve implementing changes to their product, conducting marketing campaigns, or engaging with customers to gather feedback. The actions taken should be designed to validate or invalidate the hypotheses and generate relevant data.
3. **Data:** Startups collect data from their actions and experiments to evaluate the validity of their hypotheses. This data can take various forms, including user feedback, analytics metrics, sales data, or market research findings. By analyzing the data, startups gain insights into the effectiveness of their actions and the accuracy of their hypotheses.

4. **Insights:** Based on the analysis of data, startups derive insights that inform their next steps. These insights may reveal patterns, trends, or unexpected findings that guide future decision-making and iteration. Insights help startups refine their strategies, pivot when necessary, and optimize their products or processes for better outcomes.
5. **Iterate:** The Hadi Cycle is iterative, meaning that startups continuously repeat the process of formulating hypotheses, taking action, collecting data, and deriving insights. Each cycle builds upon the previous one, allowing startups to learn and adapt over time. Through iteration, startups can refine their understanding of their target market, improve their products, and increase their chances of success.

In summary, Hadi Cycles provide startups with a systematic framework for experimentation and learning. By following this process, startups can validate their assumptions, optimize their strategies, and iterate towards achieving product-market fit and long-term success. In our team we decided to stick to this framework in order to be not only fast, but data-driven and effective at the same time.

2.4. ORGANIZATIONAL STRUCTURE

Speaking of organizational structure, we decided to operate as a limited liability company using an automated simplified accounting system registered in the Innopolis Special Economic Zone in the legislation code of the Russian Federation. We choose to have 0 employees in the company, hiring them as contractors before we show long periods of stable revenue growth.
This approach allows:

- to be very tax-efficient,
- minimize bureaucratic difficulties,

- to get state funding in the form of grants,
- get access to Innopolis infrastructure.

3. MARKET RESEARCH

The market research described in the chapter contains 3 parts.

The first part is the industry analysis, and it involves understanding the broader context in which our product operates. This includes identifying key trends, challenges, and opportunities within the industry. We also assess factors such as the regulatory environment, technological advancements, and consumer behavior that may impact our business.

The second part involves estimating the market (TAM-SAM-SOM) for our product. This includes identifying the number of potential customers or target market segments, their purchasing power, and the total revenue potential of the market. Market sizing helps to understand the scale of opportunity and make informed decisions about market entry and growth strategies.

Competitor analysis involves identifying and analyzing existing and potential competitors in the market. This includes assessing their strengths, weaknesses, market positioning, and strategies. Understanding competitor offerings, pricing, distribution channels, and customer base helps to identify gaps in the market and differentiate our product effectively.

3.1. INDUSTRY

Objective

The market research objective is to prove that the market on which a startup is operating is perspective and viable. I will provide comprehensive arguments that the retail segment of the Russian stock market is booming despite the new political and economic landscape Russia is facing after 2022.

Economical and political context

Challenges which Russia faced in 2022 are drastic, and the key factor playing now is that foreign capital has left the country. It played a significant role on the Russian stock market. In the end of 2021 according to the Moscow Exchange (MOEX) data volume breakdown by client types in Russian equities was following:

Foreign Investors	**Russian retail investors**	**Russian banks and brokers**	**Russian companies and funds**
48%	40%	9%	3%

Foreign investors held the biggest stake, and now it is not the case anymore. Public media consensus was that Russian stock market was doomed to degrade after 2022 combining two factors:

- The leave of foreign capital;
- Depression of local investors.

Condition of the stock market after 2022

However, it didn’t happen and trading volumes of Russian stock market continued to grow reaching all-time record high 7.5 trln. ₽ in 2023, as if no foreign investor left the market.

Year	2017	2019	2021	2023
Volume	3.5 trln. ₽	3 trln. ₽	4.2 trln. ₽	7.5 trln. ₽

The number of unique clients on MOEX continued to boom reaching 25.1 mln. in December 2023:

Year	2017	2019	2021	2023
Number	1.3 mln.	3.9 mln.	16.2 mln.	25.1 mln.

According to Google Trends investment popularity in Russia also peaked:

Year	2017	2019	2021	2023
Relative Number	8	12	38	41

Projection on the future of the Russian stock market by MOEX

According to the MOEX following factors are going to play the key role in forthcoming years:

- Retail investor is becoming a key source of demand on the equities market
- The need for attracting shareholder capital will continue to grow.
- The necessity of promoting shareholder activism to prevent the degradation of corporate culture among Russian issuers.

Projection on the future of the Russian stock market by the central bank of the Russian Federation

The development goals of the financial market, as defined in the "Main Directions":

1. Development of a modern financial market to meet the needs of the Russian economy in investments for structural transformation, as well as in efficient payment mechanisms.
2. Strengthening the trust of retail consumers and retail investors in the financial market by enhancing its security, increasing financial literacy, and expanding the availability of financial services to citizens, meeting their needs for financial services.
3. Ensuring financial stability - uninterrupted functioning of the financial market, including under stress conditions.

The potential of the Russian stock market

The above statements prove that the Russian stock market is rapidly growing and has the support by the government. However, it does not mean that it has further potential of growth. In order to showcase the potential let’s compare it with the stock market of the United States.

Market	Own stocks	Own personal retirement investment account	Capitalization to GDP ratio
Russia	3%	3,5%	36%
USA	60%	40%	173%

The above numbers prove that the Russian market has at least 10-fold potential of growth.

Conclusions:

- The Russian stock market has been growing during last years;
- The Russian stock market has at least 10-fold potential of further growth;
- The retail investor on the Russian stock market will continue to play a huge role, and to take this audience as a focus is a good play for a startup.

3.2. MARKET SIZE

Method

TAM SAM SOM top-down method is used in determining size of the market. It is not the only way of doing it, but this method is familiar to most investors, thus it is one of the most effective, if we want to address our point.

The total available market (TAM)

It includes all people in the world that can potentially be our customers. TAM identifies the largest possible amount of revenue we can bring in by selling our product or service across the globe. We use TAM to address our startup's potential for growth.

According to a 2022 report by the World Bank, there is a wide range of stock market ownership, with some countries having ownership rates as high as 80% and others as low as 1%. However, the global average for stock market ownership is 23%. In January 2022, the world population was estimated to be around 7.9 billion people, which gives us 1,817 billion of investors in stocks in the world.

If every single one of them becomes our subscriber bringing 500₽ monthly, it will result in 908 billion ₽ of monthly revenue. It indicates that the available market is huge and more than enough for a venture.

The serviceable available market (SAM)

It refers to the portion of TAM I can reasonably access based on my geographical reach, regulatory requirements, or pricing/quality requirements. SAM reflects the real market I can embrace with my product.

I consider entering other markets after taking a significant portion of the Russian stock market, however, it is the goal of future periods. Assessing SAM I prefer to address the Russian stock market with its size of 30 million retail investors owning 53,3 million brokerage accounts according to the January 2024 report by MOEX.

If every single one of them becomes our subscriber bringing 500₽ monthly, it will result in 15 billion ₽ of monthly revenue.

The serviceable obtainable market (SOM)

SOM identifies the portion of SAM I can actually service, considering that I have competitors and can’t make the entire market consume only my product.

Based on MOEX numbers we know that there are 30 million of retail investors on the Russian stock market. However, only around 10% of them are active and

actually making trades. Our product is aimed towards active investors, and only this 10% must be considered as obtainable clients.

On top of it, we don’t have any infrastructural competitor advantage or investment resources, thus the realistic market share of active investors we can hold is 1%, which gives us 30,000 of paying users resulting in 15,000,000 ₽ of monthly revenue or 180,000,000 ₽ of yearly revenue.

3.3. THE COMPETITION

In this chapter I will list indirect competitors of the product. In order to systematise information about them I will use SWOT-diagram. Direct competitors will not be listed, as they don’t exist for the moment.

YoloStocks | Real Time Meme Stocks Tracker

https://yolostocks.live

The website operates on the US market covering only US stocks. The functionality is very similar to what we offer to our clients, however, there is no direct competition because we operate on different markets. Direct competition is possible only in future when we enter (hypothetically) the US market.

S: monetization by donations, open methodology, API available, historical data available

W: weak web-only interface, no bot interface, no mobile app

O: more effective ways of monetization can be introduced

T: advent of competitors with much better interfaces and user experience

Brand Analytics

https://brandanalytics.ru

The company operates mostly on the Russian market offering data on mentions of any company. It is used mostly in B2B by brand managers and PR specialists, but its functionality can be used to solve the problem of our clients too. However, the pricing is too high for B2C and the company doesn't have any special offer for the client segment we aim. BrandAnalytics is also our indirect competitor – we share the same market, we offer similar functionalities, but we aim different client segments.

S: wide range of functionality, API, strong brand

W: B2B only, high price

O: they can "unbundle" their product for B2C in different sub-niches including stock market

T: they become too big and slow risking to loose market opportunities

Intel Invest

https://intelinvest.ru

The company operates on the Russian market and helps to solve the same problem retail investors have as we are trying to solve. Their method is different though. They offer intruments to track investor's portfolio and to find better stocks to invest based on fundamental analysis. Intel Invest and our product can be perfectly used in the same time by one investor. Because of this fact I see the potential to partner and collaborate with this company (to cross our marketing activities for example).

S: Relatively huge client base for a small startup, lean startup principles (small team, small expenses), modest price

W: Portfolio tracker is not very needed by investors (this assumption is based on the client interviews I did before), stock analysis by intel invest is far from the best on the market.

O: The introduction of new product features adding new subsription levels

T: abundance of direct competition to portfolio tracker and to stock analysis instrument

Black Terminal

https://blackterminal.com

Black Terminal is one of the most famous sources of market multipliers and financial data of public companies in Russia. It is far from being perfect (a lot of inaccuracies in calculations), but its product is satisfactory for some client segments. Blackterminal is our indirect competior, because they solve the same problem as we do – we both help client to find investment ideas, but we do it very differently.

S: Relatively huge client base, top of mind brand for many retail investors, modest price

W: inaccuracies in calculations

O: The instrument can be improved in terms of quality, which can result in huge leap in the size of client base

T: abundance of direct competition, the advent of the competition provided by one of huge banks or brokers

Dohod

https://www.dohod.ru

The main business of Dohod is capital management, but they have several online tools (stock screeners, dividend calendars, stock analysis tool) for retail investors, which makes them indirect competitors of our product.

S: reputation, old well-established brand, their numbers are considered to be a benchmark by many, their online tools are free to use – they serve a role of a marketing medium for the company

W: heavily regulated by central bank as a capital management company

O: they can unbundle their online tools in a separate business and start monetize it

T: advent of the competition provided by one of huge banks or brokers to their capital management business from one side and numerous small startups as a competition for their online tools from the other.

Conclusion:

It is not the full list of indirect competitors. The abundence of indirect competition means that the audience is worth to be our goal. The lack of direct competition means higher risk that our product market fit is weak, however, in the next chapter I will try to prove it is not. I expect to see the advent of direct competition in several quarters.

4. PRODUCT MARKET FIT

In the Market Research chapter we proved the Russian retail investment market to be perspective and large enough for our startup to operate on. However, the presence itself on this market is not enough to launch a successful business. In order to get more chances we need to find product-solution fit and then product-market fit.

“Product-market fit” is a term used in entrepreneurship and product development to describe the point at which a product satisfies a strong market demand. In other words, it's when a product meets the needs of its target market so well that customers are highly satisfied, leading to rapid adoption and growth.

“Product-solution fit” is a concept similar to "product-market fit" but with a narrower focus. While product-market fit evaluates how well a product satisfies the broader market demand, product-solution fit specifically assesses how well a product addresses the needs and pain points of its target customers.

To determine our “problem-solution fit”, we hire customer development methodology, which is the most straightforward way to find the “pains” of one’s target audience.

4.1. CUSTOMER DEVELOPMENT

Customer development is a methodology developed by entrepreneur and author Steve Blank, which is aimed at building and validating startup ideas by systematically engaging with customers. It is closely associated with the Lean Startup methodology and is often considered an essential component of the early stages of startup development.

Customer development methodology implies that each interview is not answering the list of specific questions. It allows the interviewer to digress from a plan based on the insights he gets during the process of an interview. However, the preplanned bone-structure of the interview helps to make it effective. It is attached in the Appendix.

The result of customer development interviews conducted by our team resulted in a following classification of retail investors, who we collaborated with:

DIY conservative investors – this category of investors are prone to long-term strategies with a little attention to short time-frame volatility and news related to stocks. Typically they incline towards FIRE movement and the idea of early retirement, but not necessarily. They more or less agree that Telegram channels heavily impact stock prices, however, they tend to think that it is not fruitful to follow these trends, and it is much more productive to stay away from these. They are not our target audience.

DIY investors / traders – people from this category understand the importance of analysis based on fundamental parameters of companies they invest in, however, they tend to use speculative market opportunities as well and are interested in stock related news and trends. They combine technical analysis, fundamental analysis, as well as, so called, behavioural analysis introduced by Sergey Popov – one of opinion leaders of retail investment market. This is our core target audience.

Capitalists – these are people looking for guidance on the stock market. They have a significant amount of capital in their disposal, but they don't work as capital managers themselves. Instead, they look for an external manager of their capital. This is very interesting audience for an entrepreneur, but the current product can't serve their needs.

4.2. PROBLEM-SOLUTION FIT

Capital Management

Based on the conducted customer development interviews we managed to make several sales to the third category of clients listed above. We got access to approximately 50 million roubles of capital allocated on 7 brokerage accounts to manage these funds. The deal is that 2 times a year we take a management fee in a form of commission 0,25% of funds themselves and 0,5% commission of success, which results approximately into 200 thousands roubles of revenue to our team. It is not a lot, but has no limit of scaling if we organize a sales team and proper marketing.

The sales themselves were not that hard to be done, and our conclusion is that there is a very high demand for capital management services on the Russian market now. We plan to develop this idea by establishing a mutual fund using the infrastructure of the already established asset management company “Record Capital”. The reason for it is to minimise starting costs of this business idea. However, the idea itself does not fit the requirements of the technological entrepreneurship program, and the key idea, which we proceeded to develop, is not this one.

Algorithmic Investment Strategies Constructor

Instead, we focused on the DIY Investors / Traders category and decided to introduce them to algorithmic investment strategies constructor, a demand for which we noticed. The reason why it’s needed is that to prosper long-term on the market you need to work using strict formalized strategy, which in the essence is the algorithm. However, it is very risky to use a strategy without

backtesting it using historical data, but the problem is that there is no simple to use product or service, which allows doing it to the retail investors.

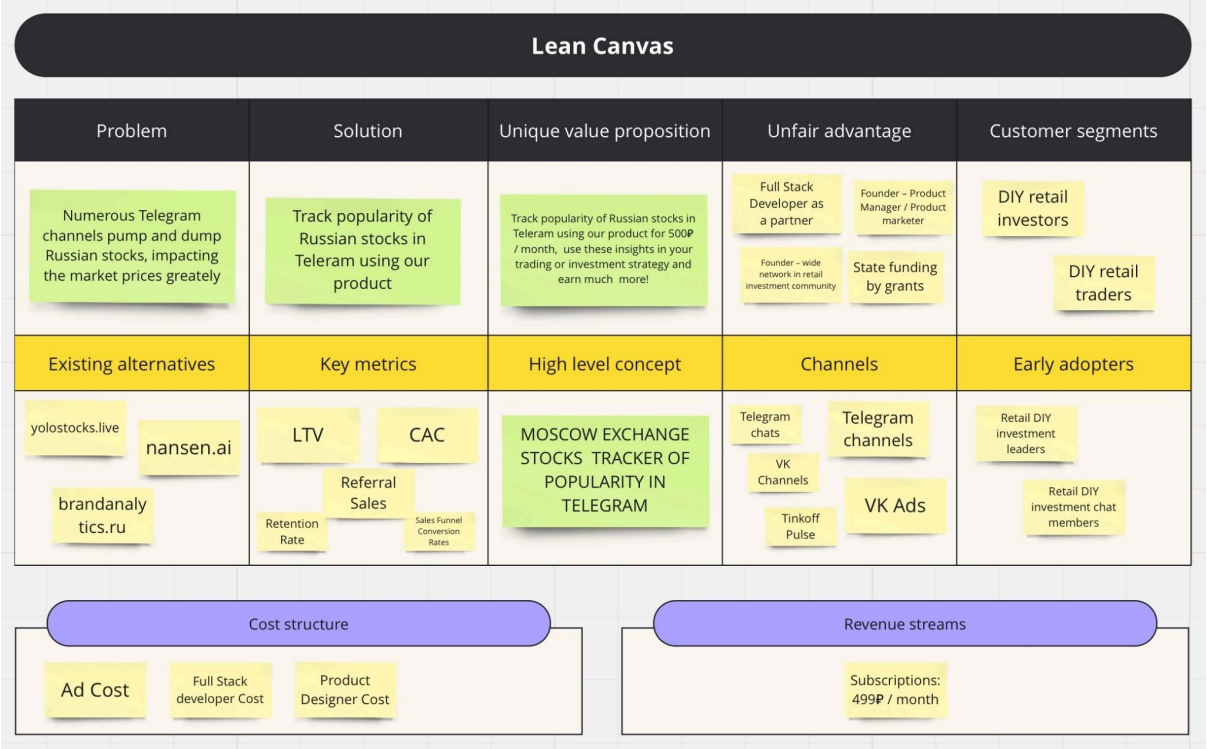
We started to build the MVP of this service, and the first step was collecting historical data of the public companies, without which the whole idea would not work. We made several attempts to solve this problem: CBonds API, RuData Interfax API, parsing, OpenAI, and even hands-on approach. Our conclusion is that it is too complicated a task technically, and we decided to leave it for future attempts, and to focus on something more realistic in production.

Russian Stocks Popularity Tracker

After our failing attempt to build the product for DIY Investors / Traders we decided to keep focus on this audience segment, and to build an MVP of another product idea, inspiration for which we received conducting customer development interviews – Russian Stocks Popularity Tracker, which is the title of this startup as a thesis. The key pain here experienced by the traders and investors is that Telegram channels heavily impact stock prices these days, but there is no instrument or product delivering a helicopter view on their activity.

4.3. LEAN CANVAS

The Lean Canvas is a one-page business planning tool developed by entrepreneur and author Ash Maurya. It is derived from Alex Osterwalder's Business Model Canvas but is specifically tailored for startups and entrepreneurs in the early stages of developing a business idea. The Lean Canvas provides a concise and visual framework for capturing and testing key elements of a business model hypothesis.



4.4. CUSTOMER JOURNEY MAP

A customer journey map is a visual representation of the steps and touchpoints that a customer goes through when interacting with a product, service, or brand. It outlines the entire end-to-end experience from the customer's perspective, starting from the initial awareness stage through to the post-purchase phase and beyond. The purpose of a customer journey map is to understand and analyze the customer's experience, identify pain points and opportunities for improvement, and ultimately enhance customer satisfaction and loyalty.



4.5. SALES FUNNEL

Speaking of Sales funnel is used in the product, the decision to use the AAARRR framework (also know as the Extended Pirate Metrics) was made. It expands upon the traditional AARRR framework by adding an additional "A" stage at the beginning. Here are the six steps of the AAARRR funnel:

Awareness: The Awareness stage focuses on building brand awareness and attracting the attention of potential customers. It involves marketing efforts such as advertising, content marketing, social media presence, and search engine optimization (SEO) to make your target audience aware of your product or service.

Acquisition: Acquisition involves acquiring new customers or users for your product or service. This stage includes activities such as lead generation, capturing contact information, driving traffic to your website or app, and encouraging sign-ups or registrations.

Activation: Activation refers to the process of converting acquired users into active and engaged users. It involves providing a seamless onboarding experience, delivering value quickly, and guiding users through key features or actions that demonstrate the value proposition of your product or service.

Retention: Retention focuses on keeping users engaged and coming back to use your product or service over time. This stage involves building strong relationships with customers, delivering ongoing value, providing excellent customer support, and implementing retention strategies to reduce churn and increase customer loyalty.

Revenue: Revenue represents the stage where users generate value for the business by making purchases or engaging in monetization activities. This includes various revenue-generating actions such as subscription payments, one-time purchases, upgrades, cross-selling, and advertising revenue.

Referral: Referral involves leveraging satisfied customers to acquire new customers through word-of-mouth recommendations and referrals. This stage focuses on encouraging existing customers to refer friends, family, or colleagues to your product or service, as well as providing a positive and shareable experience that encourages organic referrals.

The actual AAARRR sales funnel of our product is shown below:



5. IMPLEMENTATION AND RESULTS

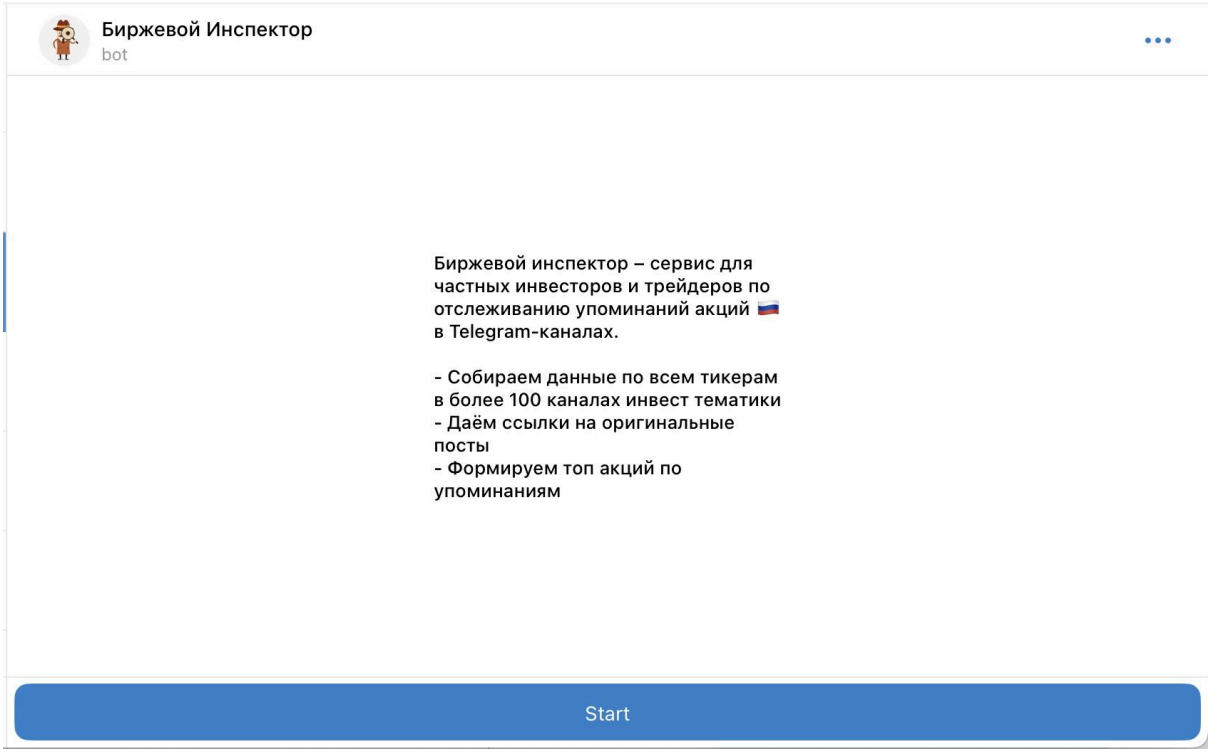
5.1. MINIMAL VIABLE PRODUCT

The chosen placement and scope of minimal viable product was the compromise between time to market and “to be” version of the product. Telegram bot format was chosen as the first platform the product is available on, because of development ease and the presence of retail investors in Telegram. Another reason is that the MVP tracks Russian stock popularity only in Telegram, which gives another reason to place the product in the Telegram bot format.

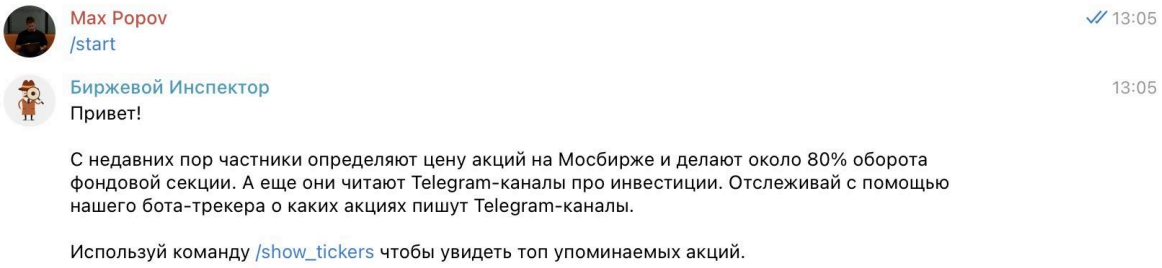
The number of features available in the MVP is, again, the compromise between time to market and the key needs of correspondents named during customer development interviews.

In this section the scope of minimal viable product will be described. The MVP itself is available via the link: https://t.me/InspectorMOEX_bot

1. The first screen a user interacts with is a starting screen in the Telegram bot. Its goal is to give a brief and concise description of the product and to urge visitors to press the “Start” button. The screenshot of this screen is given in the “Minimal Viable Product Screenshots” section of the Appendix, as well as all other screens described in this section.



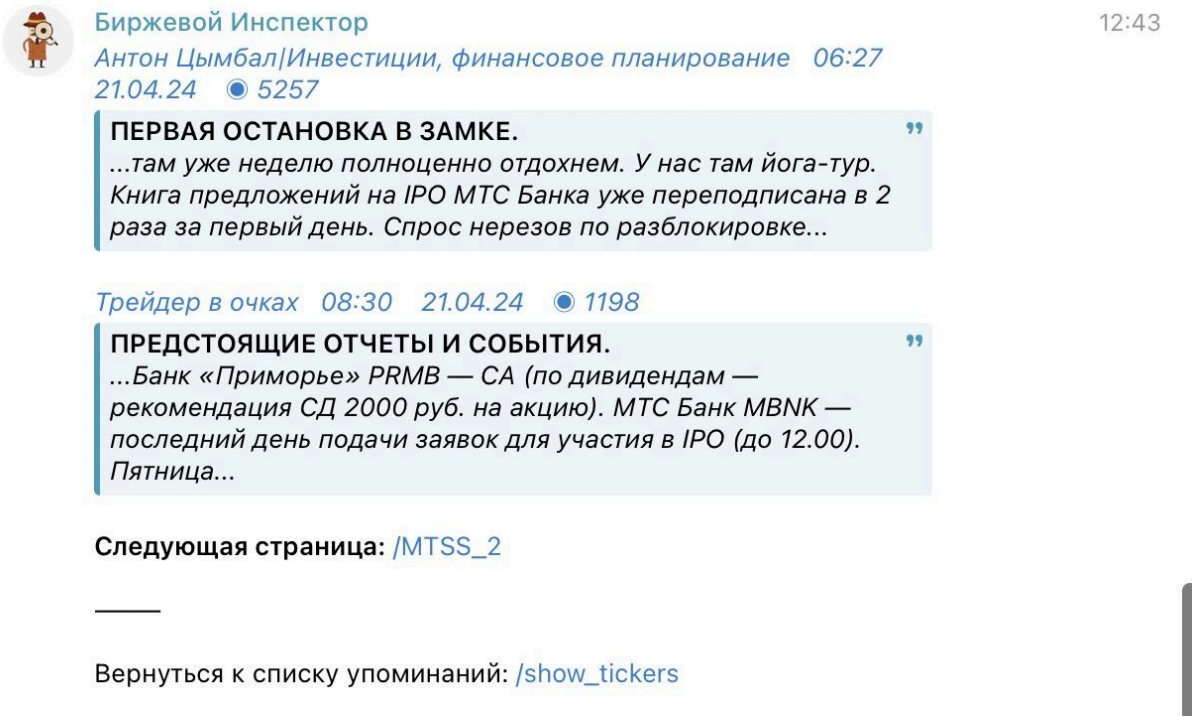
2. The second screen is a greetings screen, which gives another short description and urges visitors to use their first command /show_tickers, which gives the list of top-10 mentioned stocks in Telegram during the last 7 days.



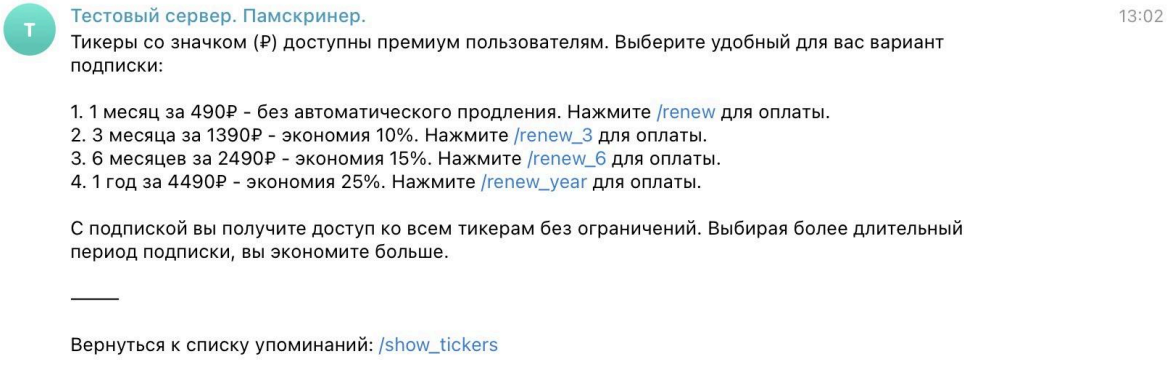
3. The third screen is the list of top-10 mentioned stocks with many options to proceed. Here is the list of all the options:
    - If visitors use one of the “stock” commands (such as /MTSS, /MOEX, /POSI etc.), they will see the list of the most popular posts related to a specific company (MTS, Moscow Exchange, Positive group accordingly);
    - If visitors use one of the “period” commands (/day, /week, /month), they will see the list of the most popular stocks during the chosen period (24 hours, 7 days, 30 days accordingly);
    - If visitors use one of the “sorting” commands (/sort_growth, /sort_volume), they will see the list of the most popular stocks sorted by mention growth speed or mention volume accordingly.
    - If visitors use one of the “list size” commands (/top10, /top25, /top50), they will see the list of the most popular stocks of a different size (10, 25, 50 accordingly).



4. Using any of the “stock” commands, visitors get to the stock page, where related to a chosen stock mentions are given. The page also has information related to a Telegram channel mentioning the stock, time of the mention, and approximate reach of the mention. Headline and the mention itself are provided too.



5. Another page worth to be mentioned is a paywall page, where visitors are informed that to get access to the information of this specific stock, they need to subscribe. They have 4 tariffs on their behalf available by 4 commands: /renew, /renew_3, /renew_6, /renew_year



6. The last page available in the MVP is the payment page leading to a Tinkoff payments service.



5.2. PRODUCT DEVELOPMENT

While choosing tools and instruments for developing the minimal viable product in our startup we were optimising “time to market”, which is one of the most important things in the methodologies we use: lean startup, bootstrapping, agile, HADI. It means that some of the chosen options were suboptimal long-term, but allowed us to move forward as fast as possible now. If the product becomes a full-scale business in future, it is quite possible that we will make the decision to recode it from scratch. However, in this section I will name tools and instruments we are using in our MVP right now.

While developing the minimal viable product in our startup Python is used as a primary programming language, as it provides convenient tools for aggregating and managing large volumes of text data. This is especially important for efficient processing and analysis of information.

For asynchronous database operations with MongoDB, pymongo and motor are utilized. Pymongo is the standard synchronous MongoDB driver for Python, while motor is its asynchronous counterpart, allowing us to leverage the benefits of asynchronous programming in Python to the fullest extent.

For interacting with users through Telegram, aiogram and telethon are used. Aiogram is a modern framework for creating asynchronous Telegram bots, providing rich functionality for message and command processing. Telethon is a library that enables asynchronous interaction with the Telegram API, including support for complex interaction scenarios such as parsing and sending data.

In addition, the MVP utilizes FastAPI for processing webhook notifications of incoming payments, running on the Uvicorn server. Uvicorn is a lightweight and

high-performance ASGI server that provides asynchronous handling of HTTP requests, supporting the high-speed operation of FastAPI. FastAPI and Uvicorn together create a powerful combination for developing asynchronous web applications, ensuring efficient management of incoming requests in real-time and supporting fast processing of payments and other operations.

All of these technologies combined make our application fast, responsive, and scalable, providing continuous and high-quality service to users.

5.3. KEY METRICS

In this section key metrics of our product are listed and explained. The importance of these metrics will be shown further in the Unit Economics section.

Click-Through Rate (CTR) is a metric used in online advertising and marketing campaigns to measure the effectiveness of an ad or a link in generating clicks from users. It is expressed as a percentage and represents the ratio of clicks on a specific ad or link to the number of times the ad or link was shown (impressions).

Cost Per Click (CPC) is a metric used in online advertising campaigns to measure the cost incurred by advertisers for each click on their ad. It represents the amount of money that an advertiser pays to the advertising platform such as Google Ads, Facebook Ads, or LinkedIn Ads (in our case VK Ads) every time a user clicks on their ad.

CR to the 1st command refers to the Conversion Rate of users who successfully complete the first command in the bot. This metric is commonly

used in chatbot analytics to measure the effectiveness of the bot in guiding users towards a specific goal or outcome right after the initial interaction.

Conversion to active user refers to the process of converting a new user or visitor into an active and engaged user of the product. In other words, it represents the transition from a passive or inactive user to one who regularly interacts with and derives value from the offering.
Conversion Rate (CR) to first sale refers to the percentage of users who make their first purchase during their interaction with the product. This metric measures the effectiveness of converting visitors into paying customers from their engagement.

Average check, also known as average order value (AOV), is a metric used in businesses to measure the average amount of money spent by customers during a single transaction or order. It provides insights into the purchasing behavior of customers and helps businesses understand the typical value of orders placed on their platform.

Customer Acquisition Cost (CAC) is a metric used in business to measure the average cost incurred by a company to acquire a new customer. It encompasses all expenses related to acquiring customers, including marketing, sales, advertising, promotions, and any other associated costs.

Retention rate is a metric used to measure the percentage of customers or users who continue to use a product or service over a specified period of time. It is a key indicator of customer loyalty and satisfaction, as well as the effectiveness of retention strategies implemented by a business. The retention rate is calculated by dividing the number of customers or users who remain engaged with the

product by the total number of customers or users at the beginning of the period, and then multiplying by 100 to express it as a percentage.

Referral Conversion Rate: This metric measures the percentage of referred leads or customers who convert into actual sales. It is calculated by dividing the number of referral sales by the total number of referred leads or customers, and then multiplying by 100 to express it as a percentage. A high referral conversion rate indicates that your referral program is effective at driving sales from referred customers.

Lifetime value (LTV) is a metric used in business to evaluate the total revenue that a customer is expected to generate over the entire duration of their relationship with a company. It is a key indicator of the long-term profitability of acquiring and retaining customers.

5.4. UNIT ECONOMICS

Russian Stocks Popularity Tracker is a SaaS. That's why its Unit Economics is built around 2 key metrics: LTV and CAC. This approach is described in the book "Unit Economics" by Daniil Khanin. The fact that predicted LTV is x1.8 times larger than CAC on 10-month timeframe means that the product is scalable, even though it's far from being super-profitable and easy to get into a cash gap. In future the team may improve the metrics included in the calculation, and the LTV/CAC ratio will increase.

Impressions	3077		**Retention Rate**	80.00%
CTR	1.00%		**2nd Sales**	0.80
Clicks	31		**3rd Sales**	0.64
CPC	80₽		**4th Sales**	0.51
CR to 1st command	65.00%		**5th Sales**	0.41
First commands	20		**6th Sales**	0.33
CR to active user	50.00%		**7th Sales**	0.26
Active users	10		**8th Sales**	0.21
CR to sale	10.00%		**9th Sales**	0.17
1st Sales	1		**10th Sales**	0.13
Av. Check	499₽		**Referral**	1
CAC	2,462₽		**LTV**	4,454₽
Active user requirements:				
# of commands	>50			
# of active days	>5			

5.5. BACKLOG OF KEY FEATURES

After the MVP launch we brought first customers to interact with the product. We contacted all the users who tried it and collected their feedback using the customer development framework again. These contacts resulted in the list of key features, which we plan to implement to make our product more satisfying to the customers. Each feature is aimed towards **the number of active users metric**, which is the main problem for our product for now.

1. The most popular feature customers requested is the historical chart of mentions for a specific company. The reason for it is that even if customers see an interesting recent abnormality in stock popularity, they want to see what was happening in the past and how it correlated with the stock price.

2. Another feature customers requested is the regular notifications from the bot, which allows to see stock popularity abnormalities regularly and assures that customers don’t miss opportunities.
3. Web or mobile version instead of the bot, which makes user interface much more comfortable for usage.

The full list of features in the backlog is much wider, however, these 3 are the key possible game changers.

6. ANALYSIS AND DISCUSSION

6.1. FINANCIAL MODEL

Relying on the unit economics numbers we provide in the previous chapter, it makes sense to project these numbers into the future and calculate the financial model to assess the overall perspective of the startup.

Before diving into the financial model, we need to reexamine the unit economics itself and try to evaluate its dynamics in future. It is never static and can be improved with time. Here is the projection of the unit economics from 2024 up to 2028.

Period	2028	2027	2026	2025	2024
Impressions	2616	2601	2671	2825	3077
CTR	0.60%	0.70%	0.80%	0.90%	1.00%
Clicks	16	18	21	25	31
CPC	100₽	95₽	90₽	85₽	80₽
CR to 1st command	65.00%	65.00%	65.00%	65.00%	65.00%
First commands	10	12	14	17	20
CR to active user	70.00%	65.00%	60.00%	55.00%	50.00%
Active users	7	8	8	9	10
CR to sale	14.00%	13.00%	12.00%	11.00%	10.00%
1st Sales	1	1	1	1	1
Av. Check	1,016₽	847₽	706₽	588₽	490₽
CAC	1,570₽	1,730₽	1,923₽	2,161₽	2,462₽
Retention Rate	88.00%	86.00%	84.00%	82.00%	80.00%
2nd Sales	0.88	0.86	0.84	0.82	0.80
3rd Sales	0.77	0.74	0.71	0.67	0.64
4th Sales	0.68	0.64	0.59	0.55	0.51
5th Sales	0.60	0.55	0.50	0.45	0.41
6th Sales	0.53	0.47	0.42	0.37	0.33
7th Sales	0.46	0.40	0.35	0.30	0.26
8th Sales	0.41	0.35	0.30	0.25	0.21
9th Sales	0.36	0.30	0.25	0.20	0.17
10th Sales	0.32	0.26	0.21	0.17	0.13
Referral	3	2.5	2	1.5	1
LTV	24,436₽	16,483₽	10,916₽	7,044₽	4,374₽

We assume that CTR will decrease because of the competition in the realm of retail investment and on the advertising platforms themselves. CPC will increase because of the same reason. Conversion to active user and to the first sale will increase because the team will learn how to improve the sales funnel. If we summarize these three factors, it will lead to the overall decrease of CAC from 2,462₽ in 2024 to 1,570₽ in 2028.

Speaking of the bottom part of the sales funnel, we assume that the retention rate will increase with time, and more importantly referral sales rate will increase from 1 to 3 (it is optimistic, but quite possible, if the customer's satisfaction will be high), which will lead to dramatic increase in LTV from 4,374₽ up to 24,436₽, which is a game changer in the startup's unit economics.

Now, we can build the financial model. We start with the 100 of paid subscribers in 2024, which is the goal. Based on the unit economics of 2024 we get the number of ad clicks, users, active users needed to achieve this goal. Also, we get the number of referral subscribers we will get by attracting 100 first sales.

	2028	2027	2026	2025	2024
Product Metrics					
Number of ad clicks	23548	18207	10684	6357	3077
Number of users	15306	11834	6944	4132	2000
Number of active users	10714	7692	4167	2273	1000
Number of paid subscribers	1500	1000	500	250	100
Number of referral subsribers	4500	2500	1000	375	100
Number of total subscribers	6000	3500	1500	625	200

These numbers allow us to calculate our revenue and ad costs. Other expenses (salaries and other costs) are projected by assuming that we continue to stick to the principles of lean startup – small efficient team in this case. This approach allows us to significant profit only in 2027-2028, balancing before on the edge of break even.

	2028	2027	2026	2025	2024
Profit and Loss					
Revenue	36,654,459₽	16,483,488₽	5,458,028₽	1,761,044₽	437,387₽
Expenses	19,954,788₽	8,529,631₽	4,361,538₽	1,640,369₽	696,154₽
Ad cost	2,354,788₽	1,729,631₽	961,538₽	540,369₽	246,154₽
Founder (Product Manager)	4,800,000₽	2,400,000₽	1,200,000₽	0₽	0₽
Partner (Developer)	4,800,000₽	2,400,000₽	1,200,000₽	600,000₽	350,000₽
Other salaries	4,000,000₽	1,000,000₽	500,000₽	250,000₽	50,000₽
Other costs	4,000,000₽	1,000,000₽	500,000₽	250,000₽	50,000₽
Operational profit	16,699,671₽	7,953,857₽	1,096,490₽	120,675₽	-258,767₽

For the the company's book value, we have the opportunity to capitalize our development costs on the balance taking into account tax-efficiency of IT companies in the Russian Federation, which is good for the company's valuation.

Speaking of the company's valuation in case of its purchase, it is always the subject of negotiations. However, we can have some orienteers.

1. Valuation of 1 yearly revenue (PS=1), which gives us approximately 36 million rubles in 2028.
2. Valuation of 4 yearly profit (PE=4), which gives us approximately 66 million rubles in 2028.
3. Valuation of 1 book value (PB=1), which gives us approximately 9 million rubles in 2028.

Book Value					
Assets	*9,350,000₽*	*4,550,000₽*	*2,150,000₽*	*950,000₽*	*350,000₽*
Company Valuation					
Value by revenue (PS 1)	*36,654,459₽*	*16,483,488₽*	*5,458,028₽*	*1,761,044₽*	*437,387₽*
Value by profit (PE 4)	*66,798,685₽*	*31,815,427₽*	*4,385,960₽*	*482,700₽*	-
Value by book value (PB 1)	*9,350,000₽*	*4,550,000₽*	*2,150,000₽*	*950,000₽*	*350,000₽*

6.2. STARTUP LIFESPAN

Relying on the financial model provided in the previous section, we can state that the startup we cover in this thesis can indeed become a very profitable small IT business with operational margin around 50%. However, there is a significant gap between potential and actual production and before it is actually made we can not ignore the factor of uncertainty. In order to save the team from the risk of getting stuck in the lossmaking venture, we agreed that if the goal of paid subscribers in 2024 will not be reached, we will close the startup and pivot to the new business idea using the same methods we describe in the chapter 2.

7. CONCLUSION

In the end of this diploma project, our endeavors have laid the foundation for a transformative journey in the realm of algorithmic retail investment. The findings and achievements resonate as significant milestones, marking the commencement of a promising trajectory for our IT-business.

Identifying a Targeted Client Segment:

An accomplishment of this study is the identification of a specific client segment within the retail investment market (DIY investors) that exhibits a profound interest in the product. Through the market research and an understanding of investor behavior, we have pinpointed a segment whose needs align seamlessly with the capabilities of our product.

Crafting Compelling Messaging and Wording:

The crafting of messaging and wording emerges as a distinctive achievement. We have formulated communication tactics that resonate with our identified client segment, eliciting positive responses and actions. The language we employ effectively conveys the value proposition, fostering understanding and engagement among potential users.

Efficient Product Delivery with Limited Resources:

Navigating the complexities of product delivery within resource constraints, we have uncovered a pathway to bring our solution to market without necessitating significant external investment. By leveraging a bootstrapping strategy and focusing on an iterative development approach, we have demonstrated the feasibility of delivering a product that meets market demand and aligns with the financial resources at our disposal.

First Sales Achievement:

A significant milestone has been achieved with the realization of our first sales. This early validation from the market underscores the resonance of our solution within the identified client segment. It serves as a testament to the effectiveness of our approach and provides an indication that our product meets a need within the retail investment landscape.

Effective Unit Economics for Scalability:

Perhaps most crucially, we have unearthed effective unit economics that lay the groundwork for scaling our product. The balance between client satisfaction, resource optimization, and profitability has been achieved. This understanding not only ensures the sustainability of our IT-business but also positions us for future growth and expansion within the competitive landscape.

BIBLIOGRAPHY / REFERENCES

APPENDIX

CUSTOMER DEVELOPMENT QUESTIONNAIRE

Demographic Information:

- Age:
- Profession:
- Marital Status:
- City of Residence:
- Income (if comfortable sharing):

Investment Habits:

- Do you invest?
- Do you trade?
- What types of investment products do you use?
- What investment instruments do you own?
- Which brokerage firms are you a client of?
- Do you use any investment services?

Experience and Usage:

- How did you start using your current investment product?
- Could you describe in detail how you started using your most recent financial product?
- What prompted you to need this product?
- How did you learn about it?
- Why did you decide to use this particular product and from this specific organization?
- How much time did you spend searching for this product?

- What is most important to you when choosing a company overall or a specific product in particular?

Information Sources:

- Where do you seek financial information?
- What books do you read on financial topics?
- What channels do you follow for financial information?
- What financial blogs do you read?

Closure:

- What else haven't I asked?
- What's important left?
- Who of your friends or colleagues can you recommend me to talk to?

PITCH DECK